

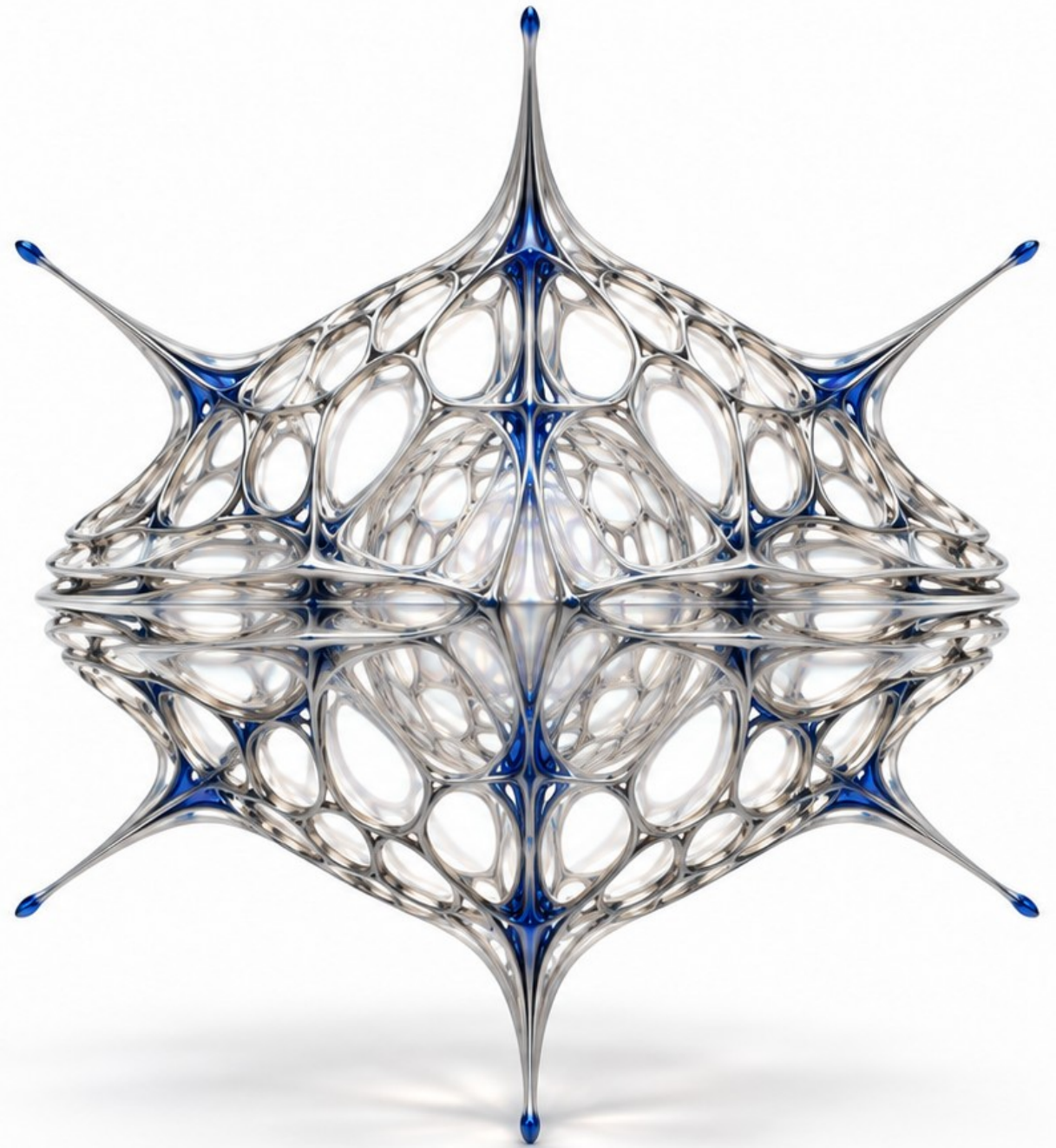
RADIOLARIA OS

Landslide Sentinel

**Local responsibility.
Replaceable intelligence.**

A cell that keeps monitoring
when the network disappears.

v0.1 / Main implementation: 15 September 2026



From research questions to executable evidence

Published challenges become explicit, inspectable software tests.

01	02	03	04
Published challenge	Software criterion	Existing kernel	Recorded evidence

Published challenge

Monitoring can fail when conditions worsen.

Measurement interval differs from delivery frequency.

Multiple surges complicate alarm duration.

Our recorded software response

C03: Usable reserve → LIMITED. No usable reserve → INSUFFICIENT.

C02: New arrival does not make an old observation current.

C07 / S5: Measured recovery supports a separate OFF action.

Synthetic observations and mock effects. Research references do not imply endorsement.

The local monitoring task

Observe, investigate, act locally, and explain what happened.

Changes in rainfall alter the next monitoring interval.

Fresh local evidence can justify a bounded signal while optional reasoning is still pending.

Recovery requires new measurements and a separate OFF decision.

Replaceable intelligence and local authority

The model is a contributor inside the cell.

Reasoning proposes a finite investigation.

The runtime binds sources, requirements and actual work.

Root remains the sole final authority; Host and Firewall enforce effects.

One account. Three honest lanes.

Each claim names how it was observed.

Lane	Actually exercised	Boundary
Controlled story	S0-S5, real core, mock effects	No paid model in that story
Live semantics	5 Gemini calls + actual Work	Synthetic site context
External readings	8 actual EA values	Historical; physical mapping unresolved

A local cell with explicit boundaries

Capabilities compose around current evidence.

01

**Typed
observations**

Time, source, units,
lineage

02

**Bounded
reasoning**

Duty-specific
proposals

03

**Current Root +
Host**

Review and admitted
dispatch

04

Results + memory

Receipts and
reusable meaning

S0-S2: build the current monitoring work

The next interval is an executed configuration.

01	S0 / Establish	Configure, intake, local memory
02	S1 / Adapt	Rainfall changes pending cadence
03	S2 / Investigate	Roles feed reviewed Work

S3-S5: act, deliver, recover

The incident ends through evidence and a new action.

01	S3 / Protect	Local ON precedes pending answer
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02	S4 / Deliver	Queue → exact acknowledgement
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03	S5 / Recover	Measured evidence → separate OFF
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What crosses into the reasoning role

A bounded view with an explicit investigative purpose.

Local and adversarial duties receive current local channels and the maintenance note.

The environmental duty receives rainfall context and its declared site/time support.

Each role receives a finite catalogue, current policy and capability snapshot.

The bounded analytical response

A useful proposal remains a proposal.

Field	Meaning	Runtime condition
diagnostic	Next bounded investigation	Allowed for this duty
source_refs	Citations into supplied context	Known, unique, nonempty
reason	Why that investigation helps	Bounded text; no executable authority

Semantic evidence packets

The runtime carries references and constraints forward.

BSEP contains the selected investigation, an input identity and contribution references.

Raw user and cross-role text are not copied into BSEP.

Candidate-only evidence still needs C, Work and Root checks.

Routing and review use the existing kernel

The domain adds meaning, not a second authority system.

01

C / Route

Current snapshot +
BSEP

02

Root review

Accepted scope and
eligibility

03

D / Review

Real bound execution
bundle

04

Work

Actual finite
diagnostic output

A proposed read becomes a reviewed read

The receipt is produced after the plan is accepted.

01

PLAN Work

Pure; no acquisition

02

Current Root

Accepted live basis

03

Native read

Packet, Host, receipt

04

RESULT Work

Consumes receipt;
Root review

Three contributions. One current composition.

All three meanings reach the real consumer.

Duty	Live selection	Actual result
Local slope analyst	Reference integrity	Current channel associations
Environmental analyst	Spatial applicability	Synthetic site identity; mapping unresolved
Adversarial reviewer	Reference integrity	Independent current-pair review

One changed question. Identical readings.

Only the source-bound maintenance note changes.

A / Current association

Is the latest reserve entry healthy?

Which current calibration is attached?

B / Historical variation

How much did reserve readings vary?

How much measured time does the window cover?

Different meaning produced different work

The difference is in execution, not just wording.

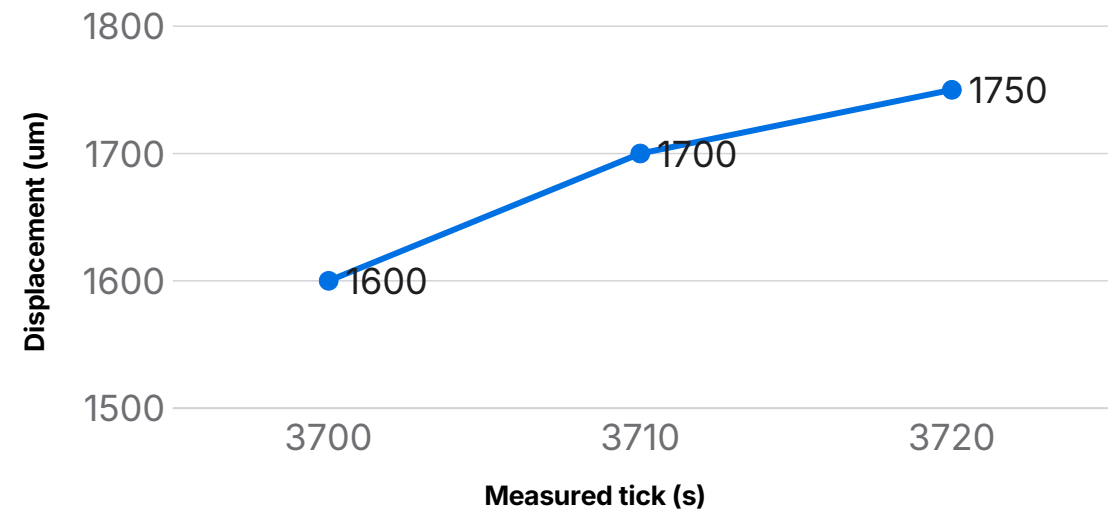
A / Current metadata

Calibration and health Distinct channel lineage

Current metadata comparison
No reserve-history read

The same-context control retained substantive work.
Both executions use the same sensor inputs.

B / Recorded mock history



3 samples / 20 s measured span / 150 um range

Recorded admitted synthetic reserve-history read
MOCK_SOURCE_READ; not physical slope measurements

A question has its own evidence needs

Source-fitness guards do not answer every question.

Reference integrity investigates current metadata relationships.

Reserve history investigates measured variation after source-fitness checks.

Evidence gaps compares supplied facts with the selected duty and intended use.

The earlier pairs remain inconclusive

A later success does not rewrite the experiment log.

01	LS2	Same work; inconclusive
02	LS2R	Overlapping purpose; inconclusive
03	LS2R2	Distinct executed investigations

E changes the next real intake

A pending configuration is not a permanent entitlement.

01

Rainfall changes

Pending packet becomes stale

02

Actual E

Changed work + retained sibling

03

Current configuration

Consumes recomputed result

04

Next intake

Uses applied cadence

The same rainfall can mean a different cadence

The accepted policy is an input, not a comment.

41

Rainfall, mm equivalent

50

**Accepted fast threshold,
mm**

60 s

Actual next cadence

Configured at tick 90. Next intake at tick 150.

Local action precedes the cloud answer

The pending worker does not own the Host.

01	Pending request	Old immutable context
02	Fresh local evidence	New bounded invocation
03	Native mock ON	Current Root and Host
04	Late response	Old identity; no automatic effect

Missing evidence changes the answer

The runtime preserves insufficiency instead of inventing confidence.

Freshness uses measurement time; receipt time cannot renew an event.

Primary missing: LIMITED. Primary and reserve missing: INSUFFICIENT.

Independent lineage, order, units and explicit intervals are checked.

OFF needs its own evidence

Three fresh pairs must cover the recovery interval.

2 channels

**Independent current
witnesses**

3 pairs

Minimum new samples

30 s

**Measured span on each
channel**

Maximum gap: 15 s. Local age: current accepted policy.

A report is immutable. Delivery is a state.

No acknowledgement means UNKNOWN, not success.

01

**Immutable
payload**

Message ID + byte
hash

02

Queued attempt

Current bounded
dispatch

03

Receiver check

Exact identity and
bytes

04

Acknowledgement

RECEIVED or
UNKNOWN

One cell, two reasoning profiles

The intended role and actual engine stay explicit.

Profile	Observed implementation	Capability boundary
Algorithms only	Fresh local work + LocalDRS	0 model transports
Internal analyst	Gemini emulates intended local SLM	Cloud origin is disclosed
Future physical SLM	Architectural substitution point	No hardware performance claim

Memory and current eligibility

LocalDRS does not resurrect an old permission.

Cold lookup, reviewed write, fresh query and bounded descent are separate steps.

Current Root, policy, calibration and dependency scope govern reuse.

Informational reuse yields new Work; it does not reopen an action packet.

Eight real values. All zero.

Zero is a measurement; an empty list is not.

8

**Actual normalized
readings**

0

Value of every reading

4163

Pinned EA source

Interval support unconfirmed. Physical site mapping not established.

Fresh transport does not make data applicable

Time support and physical meaning are different questions.

Question	Checked relationship	Honest possible result
Temporal	Interval + statistic + age	Supported, missing, incompatible, historical
Synthetic scope	Source site versus target site	Match or mismatch
Physical relevance	Verified gauge-to-slope mapping	Not established

Five ways to challenge the interpretation

The checks exercise consumer behavior.

Contrast	Challenge	Observed boundary
C01	Change only the note	Different substantive work
C02	Old event; new receipt	Current applicability check
C03	Remove local witnesses	LIMITED / INSUFFICIENT
C04	Change chronology or lineage	Typed evidence checks
C05	Reduce Work budget	2 → 1 → infeasible 0

Five ways to challenge continuation

A prior success does not authorize the next action.

Contrast	Challenge	Observed boundary
C06	Optional provider still pending	Local ON first
C07	Apparent recovery	Fresh measured window + OFF
C08	Different local channel types	Same site Root
C09	Delivery and replay ambiguity	Exact acknowledgement; no permission reuse
C10	Policy/calibration changes	Current E and memory eligibility

An inspectable public derivative

The evidence map is broader than the frozen seal.

Public safe package: exact LS2R2 fields in a frozen projection, manifest and publication.

Whole-story claim map: source archive/member/hash links with historical scope.

Presentation and XML: new explanatory derivatives with their own identities.

Independent pin verification

Verification consumes saved evidence, not new model calls.

01

External pin

Independent
expected identity

02

Reopen bytes

Manifest and
publication

03

Check relations

Supported safe
projection

04

PASS / refusal

No new external
execution

Minutes where work is actually performed

The timings describe separate observed scopes.

Observed scope	Elapsed	Interpretation
Latest live stage	205.70 s	Includes 5 Gemini calls and 1 D
Its common D	100.55 s	Subset of that stage
Its Gemini calls	67.42 s	Sum of the 5 provider durations
Earlier policy E	242.59 s	Different recorded experiment
Earlier policy stage	381.19 s	Includes D/E and checks

The implementation is in main

Commit 2f328be binds the landed demonstration.

46 landed files; 1,015 protected base files preserved.

Admission, 9 Living registrations, release binder and replay passed.

Source commit 2f328be; ordinary push; clean owner tree.

A demonstrated coordination primitive

Meaning influences work. Current authority controls action.

Established here

Live meaning changes actual work

Local current action survives optional delay

Evidence remains inspectable and scoped

Requires future validation

Physical sensing and site mapping

Installed local model and hardware timing

Operational deployment guarantees

From domain requirements to testable behavior

A reusable foundation with inspectable decisions and results.

Published question → explicit requirement → executed example → evidence.

Approximately 12 hours on the existing kernel: owner-reported implementation time.

Human direction, prepared AI assistance and Codex. Prior kernel development is separate.